



Applications of NFC during A Little World project
(Courtesy: NXP Semiconductors)

photo frame, or print them directly by placing it near the NFC-enabled printer. Setting up wireless home office network is also possible by using NFC mobile phone.

"NFC is not meant for bulk data transfer. The data dump therefore is not very high. If you want to do video streaming or music streaming you will have to use some other mode like Bluetooth," says Ashok Chandak, senior director-global sales and marketing, NXP Semiconductors.

But setting up Bluetooth connection is a cumbersome process. However, you can initiate a connection using NFC and then transfer data via Bluetooth. While NFC can ease the connection, Bluetooth can provide higher data transfer rate.

"During an NFC handshake, a device can detect Bluetooth to synchronise, as the standard is defined in NFC as well as in Bluetooth. The connection can be set up through NFC, and information can be passed from the NFC software to Bluetooth software. The Bluetooth Forum has the information on how to receive the connection information when the connection is initiated," explains Kandasamy.

For Bluetooth 2.0, you can get speed of up to 3 MBps as compared to NFC which offers only up to 424 kbps. "So, there is a connection speed hierarchy. It starts with NFC, which is secure and easy to establish, then comes Bluetooth with medium speed, and finally we have WLAN, which gives the highest data rate," he adds.

NFC for secure transaction.

The most useful feature of NFC is its secure application in combination with smart card. NFC enables users to make payments through its point-of-sale (POS) application. A mobile phone can be used as a

transport card, debit card, credit card, discount card, and behave as the all-in-one financial services device. Thus, these types of applications require the highest level of security.

The banks offering transaction services via NFC transfer credit card details onto a secure element in mobile phone. You can store information of multiple credit cards on the same device. There is an agreement between the bank and the service provider for all the communications, which are

further encrypted. Mobile payment offers expanded payment functionality, increased convenience, enhanced security and faster transactions. Hence, the inherent security feature of NFC-enabled mobile phones is a powerful tool for payment and financial services.

"NFC phone allows instant transfer of money, irrespective of recipient's location," says Khan.

"The same can be done using SMS banking services. However, you cannot use the money received through SMS right away. But with NFC, the recipient can use the money right away by using the NFC phone at NFC-enabled merchants, ATMs, or further transfer the money to another person's NFC phone," he adds.

NFC credit card payment or debit card use is also programmed for access using personal identification number (PIN). When you bring your mobile phone in the proximity of a reader, it detects the device, but will debit the amount only when you enter a valid PIN.

NFC hardware and software

In an NFC-enabled mobile phone, the most important component is the NFC controller IC with an antenna for RF

What is NFC Forum?

NFC Forum was co-founded by NXP, Sony and Nokia in 2004 to promote the NFC technology. It works jointly with other 140 members to develop specifications and standards and ensure inter-operability between various manufacturers. The forum, which comprises chip manufacturers, telecom operators, OEMs, applications developers and financial service providers, aims to promote the NFC technology in consumer electronics, mobile phones and PCs.

Innovation Research and Technology, Inside Contactless, Mastercard Worldwide, Microsoft Corp, NEC Corp, NTT Docomo, Panasonic, Renesas Technologies, Samsung Electronics, Visa, are some of the members of the NFC Forum.